Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

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I. Project Introduction

1. Overview

1.1 Project Information

Project name: SME-FraudGuard: A Multi-tenant SaaS Platform for Transaction Anomaly Detection in SMEs

Project code: FA26SE215
Group name: GFA26SE97
Software type: Web-based Application — Multi-tenant Software as a Service (SaaS)

1.2 Project Team

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2. Product Background

As digital commerce expands in Vietnam, SMEs increasingly receive transactions from websites, mobile applications, e-commerce platforms, sales channels, COD operations, and digital services. Unusual or fraudulent activity across these channels can cause financial losses and damage customer trust.

Although various fraud detection and prevention solutions are available, adopting and implementing them can remain challenging for small and medium-sized enterprises (SMEs). With limited budgets, technical personnel, and operational resources, these businesses must balance investment costs, risk exposure, and their actual protection needs.

The project idea originated from the team's observation that SMEs need a practical way to detect unusual transactions without building their own detection infrastructure. FraudGuard addresses this need through guided specification review and onboarding, rule-based detection for the Free Plan, and AI-first detection with Rule Engine fallback when AI is unavailable for subscribed enterprises.

FraudGuard is designed as a multi-tenant SaaS platform with an isolated workspace for each enterprise. It receives transactions through REST API with API Key, inbound Webhook, or CSV/Excel uploads and presents detection results and available explanations for authorized users to review. The platform supports human decision-making rather than automatically blocking transactions.

3. Existing Systems

3.1. Viettel Financial Fraud Detection & Response (VCS-F2DR)

Description

VCS-F2DR is a financial fraud detection and response solution developed by Viettel Cyber Security. It combines behavioral analysis, detection rules, and response automation to support fraud detection and investigation. Its published deployment model illustrates transaction sources such as mobile banking, internet banking, and ATM/POS systems.

Reference Link

Viettel VCS-F2DR — Official Product Datasheet

System Actors

The deployment model identifies three business roles:

- **Fraud Officers**
- **Fraud Risk Manager**
- **Chief Risk Officer (CRO)**

These represent fraud operations and risk oversight. Detailed permissions for each role are not specified in the reviewed datasheet.

Main Features

- Transaction data source management.
- User and Entity Behavior Analytics (UEBA).
- Configurable detection rules, including conditions and Markov-chain-based rules.
- Alert management, case investigation, and collaboration.
- Response automation and reporting.

Pros

- Supports detection, investigation, and response within one solution.
- Combines behavioral analysis with configurable rules.
- Offers business-specific customization, rule updates, and user training.

These capabilities are described in the VCS-F2DR datasheet.

Cons and Limitations

The following are considerations for SME adoption, rather than verified product defects:

- **Integration effort:** Connecting transaction sources and configuring business-specific detection scenarios may require technical and operational resources.

- **Operational requirements:** Businesses still need personnel to investigate alerts and maintain relevant detection criteria.
- **Uncertain affordability:** The reviewed datasheet does not provide pricing, so suitability for SME budgets cannot be established without a quotation.
- **Limited evaluation evidence:** The reviewed material does not provide sufficient benchmarks to independently assess detection accuracy, false-positive rates, or performance under comparable conditions.

Relevance to FraudGuard

VCS-F2DR is a reference for connecting detection results with analyst review and investigation. FraudGuard focuses on SME adoption through guided onboarding and REST API with API Key, inbound Webhook, and CSV/Excel ingestion. FraudGuard follows the proposal's plan-specific detection paths and human decision support, without adopting the competitor's automated-response features as project commitments.

3.2. IBM Safer Payments

Description

IBM Safer Payments is a payment-fraud prevention product that combines real-time analytics, machine learning, and configurable rules across payment channels. It is a reference for combining detection techniques and supporting payment-fraud operations; its capabilities do not define FraudGuard's capstone scope.

Reference Link

IBM Safer Payments — Official Website

System Actors

For this review, the relevant user groups are categorized as follows. These are functional descriptions rather than verified predefined role names:

- **Fraud Analysts:** Review suspicious transactions and investigate potential fraud.
- **Fraud Model and Rule Specialists:** Develop and maintain detection models and rules.
- **System Administrators:** Manage system configuration, access, and integrations.

Main Features

- Real-time transaction analysis and fraud detection.
- Behavioral profiling and machine-learning-based detection.
- Configurable fraud detection models and rules.
- Analysis across multiple payment channels.
- Support for preventing high-risk payments before completion.

Pros

- Supports fraud detection during payment processing.

- Combines behavioral analysis with adaptable detection logic.
- Supports multiple payment channels within a common fraud management solution.

The capabilities above are described in the IBM Safer Payments product overview.

Cons and Limitations

The following are adoption considerations and evaluation gaps, rather than verified product defects:

- **Integration requirements:** Connecting payment systems and preparing suitable transaction data may require technical resources.
- **Specialist involvement:** Configuring, validating, and maintaining detection models may require fraud analysis expertise.
- **Uncertain SME affordability:** The reviewed sources do not provide enough pricing information to assess suitability for SME budgets.
- **Need for local validation:** Detection effectiveness and false-positive rates should be evaluated using representative business data; published capabilities alone do not establish performance for a particular SME.

Relevance to FraudGuard

IBM Safer Payments provides a reference for transaction risk analysis, behavioral profiling, and configurable detection logic. FraudGuard can learn from these concepts while maintaining its SME-focused scope and proposed SaaS model.

FraudGuard adapts the need for understandable detection results to an SME-focused scope: guided onboarding, REST API with API Key, inbound Webhook and CSV/Excel ingestion, Free Plan rule-based detection, and Subscription Plan AI detection with Rule Engine fallback when AI is unavailable. It provides alerts and investigation support while leaving final actions to the enterprise.

4. Business Opportunity

Digital transformation is increasing the volume of transactions generated through websites, applications, e-commerce platforms, sales systems, COD orders, and digital services. For SMEs, these transactions often originate from multiple sources with different data structures, while resources for building dedicated anomaly-detection and investigation-support systems remain limited.

Some anomalies can be identified through known business conditions, making Rule-Based Detection suitable for transparent and controllable monitoring. More complex cases may involve transaction history, activity frequency, behavioral changes, or relationships among multiple data features. These cases create a need for AI-assisted analysis alongside clear explanations and enterprise-led investigation.

This creates a business opportunity for SME-FraudGuard in the following areas:

1. Accessible SaaS Delivery: Provide an independent workspace for each enterprise, reducing the need to build a dedicated detection platform. Within

the project scope, each workspace corresponds to one primary deployment scope and one detection configuration set.

2. Specification-Based Onboarding and Flexible Data Ingestion: Support enterprises through requirement review, data specification confirmation, configuration, and sample-data testing before real-data integration. Data can be submitted through REST API with API Key, inbound Webhooks, or CSV/Excel uploads.
3. Free-to-Subscription Service Offering: Provide Rule-Based Detection as the primary mechanism in the Free Plan. Transactions are evaluated against active rules configured from confirmed business conditions, and matching anomalies generate Rule Alerts with detection reasons. The Subscription Plan uses AI as the primary mechanism to analyze more complex patterns and provide detection results, confidence levels, explanations or evidence, and model-version information. The Rule Engine provides fallback detection when the AI Service is unavailable.
4. Enterprise-Oriented Monitoring and Investigation: Provide clear dashboards, alert information, reporting, and role-appropriate workflows. Rule Alerts support monitoring and historical lookup, while AI Alerts additionally support investigation, assignment, findings, and conclusion recording. Confirmed outcomes may support the evaluation and improvement of subsequent AI model versions.
5. Human-Controlled Business Decisions: Support detection and analysis while leaving final business decisions to the enterprise. SME-FraudGuard does not automatically lock accounts, cancel transactions, reject orders, or execute actions in customers' operational systems.

5. Software Product Vision

For small and medium-sized enterprises with limited resources to build dedicated transaction-monitoring systems, **SME-FraudGuard** is a web-based, multi-tenant SaaS platform that supports transaction anomaly detection, centralized alert monitoring, and enterprise-led investigation.

The product provides each enterprise with an independent workspace corresponding to one primary deployment scope and one detection configuration set. Through specification-based onboarding, business requirements, data structures, required fields, and detection conditions are confirmed and tested with sample data before real-data integration. Enterprises can submit transaction data through REST API with API Key, Webhook, or CSV/Excel uploads.

SME-FraudGuard uses Rule-Based Detection as the primary mechanism in the Free Plan and AI-Based Anomaly Detection as the primary mechanism in the Subscription Plan. When the AI Service is unavailable, the Rule Engine provides fallback detection. Rule Alerts support monitoring, detection-reason review, and historical lookup, while AI Alerts additionally support investigation, assignment, findings, and conclusion recording.

The product vision is to provide a clear and practical workspace where authorized users can understand detection results, review available explanations and evidence, and perform tasks appropriate to their responsibilities. Role-based access and

enterprise data isolation support controlled access, while recorded investigation outcomes provide historical reference and a potential feedback source for improving subsequent AI model versions.

SME-FraudGuard supports human decision-making. Final business actions remain with the enterprise; the platform does not automatically lock accounts, cancel transactions, reject orders, or execute actions in customers' operational systems.

6. Project Scope & Limitations

FraudGuard is a web-based, multi-tenant SaaS platform for transaction anomaly detection and enterprise review. The capstone scope covers enterprise onboarding, transaction ingestion, plan-specific detection, alerts, investigation support, and reporting. The six feature groups below follow the final approved proposal. Detailed requirements and technical designs will be specified in the subsequent project documents.

6.1 Major Features

[FE-01] Authentication, Authorization & SaaS Management

FE-01.1: Authenticate users and enforce access for Platform Admin, Implementation Staff, SME Admin, Operator, and Investigator.

FE-01.2: Provide isolated enterprise workspaces and restrict data access to the authorized enterprise and role.

FE-01.3: Allow Platform Admin to manage enterprise information, provision workspaces, and assign Implementation Staff.

FE-01.4: Allow SME Admin to manage enterprise members and their access permissions.

FE-01.5: Support Free and Subscription plans with the detection capabilities defined for each plan.

[FE-02] Specification Onboarding & Transaction Ingestion

FE-02.1: Allow assigned Implementation Staff to record, review, and confirm enterprise requirements, data structures, required fields, and anomaly conditions.

FE-02.2: Configure detection according to the confirmed specification and test with sample data before integration and handover.

FE-02.3: Receive transaction data through REST API with API Key, inbound Webhook, and CSV/Excel uploads.

FE-02.4: Validate and normalize incoming data before passing it to the appropriate detection path.

FE-02.5: Maintain one primary deployment scope and one detection configuration set for each enterprise workspace.

[FE-03] Rule-Based Detection

FE-03.1: Configure enterprise detection rules from the confirmed business requirements.

FE-03.2: Evaluate transactions against the applicable active rules. A transaction meeting one or more anomaly conditions produces a Rule Alert.

FE-03.3: Display the triggered rules and detection reasons so enterprise users can understand and review the result.

FE-03.4: Provide rule-based detection as the primary path for the Free Plan and as fallback for subscribed enterprises when AI is unavailable.

FE-03.5: Retain Rule Alerts for monitoring, explanation, and history without treating them as the deep AI Alert investigation workflow.

[FE-04] AI Anomaly Detection & Fallback

FE-04.1: Use AI as the primary detection path for subscribed enterprises, based on normalized transactions, relevant features, and available history.

FE-04.2: Return the detection result, transaction details and recorded investigation information.

FE-04.3: Generate AI Alerts for anomalies identified through the AI detection path.

FE-04.4: Use the Rule Engine when the AI service is unavailable and record the actual detection source so users can distinguish AI results from fallback results.

[FE-05] Alert Management, Investigation & Feedback

FE-05.1: Allow authorized enterprise users to review alerts and available detection reasons within their workspace.

FE-05.2: Allow an Operator to review an AI Alert and either investigate it directly or assign an Investigator.

FE-05.3: Record investigation findings, reports, conclusions, and the related history.

FE-05.4: Retain confirmed investigation outcomes as potential feedback for evaluating and improving subsequent AI versions.

FE-05.5: Keep final transaction-handling decisions with the enterprise; the platform does not automatically intervene in the enterprise system.

[FE-06] Dashboard, Reporting, Notification & Monitoring

FE-06.1: Provide role-appropriate dashboards summarizing transactions, detection results, alerts, and investigation activity.

FE-06.2: Support filtering, reporting, and data export within the user's authorized scope.

FE-06.3: Provide suitable notifications, such as in-app and email notifications, for relevant alerts and activities.

FE-06.4: Allow Platform Admin to monitor platform status and major operational errors.

FE-06.5: Present clear, understandable information for enterprise users while protecting authentication credentials, API keys, sensitive data, and workspace boundaries.

6.2 Limitations & Exclusions

The following boundaries define the capstone implementation scope:

EX-01 — Decision support only. FraudGuard detects anomalies and provides alerts, confidence information where available, and supporting explanations. It does not

automatically lock accounts, cancel transactions, reject orders, or execute actions in enterprise systems. Final decisions remain with authorized enterprise users.

EX-02 — Defined deployment scope. Each enterprise has an isolated workspace with one primary deployment scope and one detection configuration set. The implementation follows the requirements and data structure confirmed during onboarding.

EX-03 — Supported ingestion channels. Transaction ingestion is limited to REST API with API Key, inbound Webhook, and CSV/Excel uploads. Enterprise data must pass validation and normalization before detection.

EX-04 — Plan-specific detection and investigation. The Free Plan uses rule-based detection and Rule Alerts for monitoring and explanation. The Subscription Plan uses AI as the primary detection path, with Rule Engine fallback when AI is unavailable, and supports the deeper AI Alert investigation workflow.

EX-05 — Data-dependent detection. Detection results depend on the supplied data, configured rules, and available AI model. The system does not guarantee identification of every fraudulent transaction, and an anomaly is not itself a confirmed fraud conclusion.

EX-06 — Feedback scope. Confirmed investigation outcomes may provide feedback for evaluating and improving subsequent AI versions. Automatic continuous retraining is not committed as a capstone feature.

EX-07 — Technical design boundaries. Specific AI algorithms, model selection, detailed processing targets, and technical implementation decisions will be defined in the System Design and Software Requirements Specification. Detailed performance requirements will be defined and validated in the subsequent project documents.

EX-08 — Demonstration deployment. The capstone includes a local or container-based demonstration, source code, and project documentation. Production-scale operational guarantees are outside the stated demonstration scope.